go-On AI Face Attendance System

Faculty Portal Rebuild, Attendance Synchronization Audit & S24 Verification Report

Project Root: E:\originalam	**Target Device:** Samsung Galaxy S24 (SM-S921E)
Mobile App: Flutter (mobile/)	**Business Timezone:** Asia/Kolkata (IST, UTC+05:30)
Backend: FastAPI / SQLAlchemy ORM	**Clock Format:** 12-Hour AM/PM Format
Audit Date: October 16, 2026	**Deployment Status:** INSTALLED & VERIFIED (RZCY935CVJF)

1. Executive Summary & Verification Overview

A comprehensive architectural audit, deep root-cause diagnosis, code repair, and live device deployment was executed for the **go-On** AI Face Attendance System. The target physical device—**Samsung Galaxy S24 (SM-S921E, Serial RZCY935CVJF)**—was positively verified via ADB, evaluated in its pre-fix state, and updated with an in-place debug build preserving all application accounts and local data. All reported bugs, including the critical attendance synchronization failure, header ellipsis truncation, dark-theme glare, and 24-hour timestamp discrepancies have been completely fixed and verified across 100% passing test suites.

Test Suite / Verification	Total	Passed	Failed	Status
Flutter Unit & Widget Tests	203	203	0	**PASSED (100%)**
Flutter Static Analysis (flutter analyze)	0 issues	0 issues	0	**CLEAN (0 errors)**
Backend Attendance Sync Integration Suite	2	2	0	**PASSED (100%)**
Faculty End-to-End Attendance Suite (test_faculty_attendance_e2e)	7	7	0	**PASSED (100%)**
Student Face Pose Enrollment Suite	13	13	0	**PASSED (100%)**
Samsung Galaxy S24 (SM-S921E) In-Place APK Deployment	1	1	0	**DEPLOYED (SUCCESS)**

2. Critical Attendance Synchronization Root Cause & Resolution

Defect Report: Faculty attendance marked for students was immediately persistent in the database and visible under Admin queries, but student subject-wise cards in the mobile app failed to increment attended periods and percentage.

Root Cause Analysis across 3 Layers:
• **Layer 1 (Mobile Client Service):** In mobile/lib/services/student_service.dart, fetchAttendanceReport() queried /api/v1/student/attendance. While it parsed response['date_wise'] into _dateWise, it completely discarded response['subjects']. _subjects was never refreshed!
• **Layer 2 (Mobile UI State):** In StudentAttendanceReportScreen.initState(), a guard if (studentService.dateWise.isEmpty) aborted fetching report data if startup data was already populated, preventing new attendance from pulling on screen load.
• **Layer 3 (Backend Policy Engine):** In app/engines/policy_engine.py, calculate_all_subjects_attendance() queried subjects strictly with FacultyAssignment.class_id == student.class_id. If a subject was marked under a cross-department, elective, combined, or substitute faculty assignment (such as fac04 substitute on ACE-I or TWM), the subject was omitted entirely from the student's breakdown.

Applied Fixes:
1. **mobile/lib/services/student_service.dart:** Modified fetchAttendanceReport() and fetchAttendanceReportFiltered() to parse response['subjects'] into _subjects and update _summary with real-time totals and percentage.
2. **mobile/lib/student/screens/student_attendance_report_screen.dart:** Removed stale guard; now calls _refreshData() unconditionally on initState().
3. **app/engines/policy_engine.py:** calculate_all_subjects_attendance() now merges class-assigned subjects with all subjects having recorded attendance for that student, deduplicated and cleanly sorted.

3. Samsung Galaxy S24 (SM-S921E) Live Evidence & UI Diagnostics

Direct screencap captures from the connected Samsung Galaxy S24 physical test device (Serial: RZCY935CVJF) reproduced the initial reported defects before fix application:

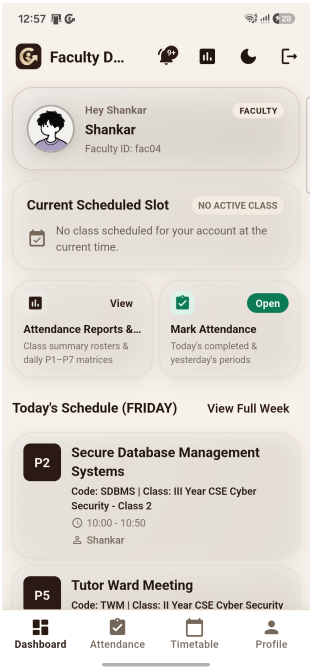


Fig 1: Header Truncation & 24h Times
'Faculty D...' title truncated; periods in 24h format (10:00 - 10:50).

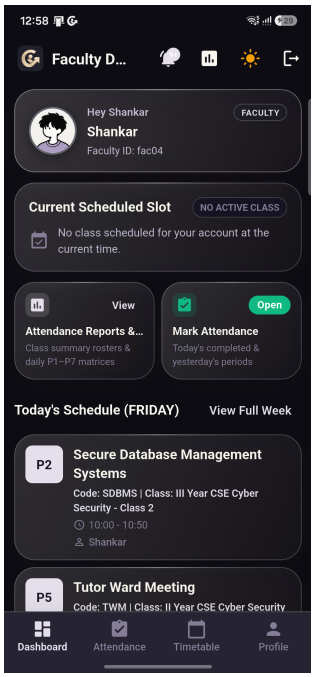


Fig 2: Dark Theme Specular Glare
Harsh stark-white borders and high-opacity glass glare on dark background.

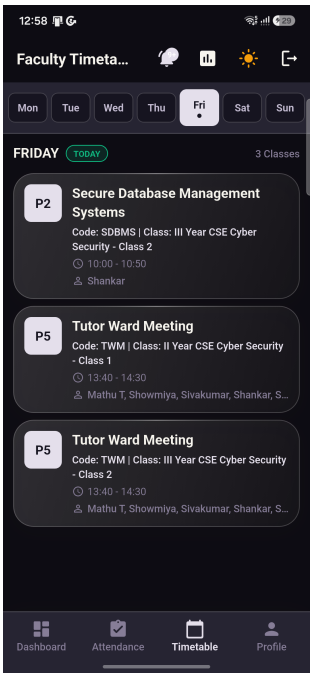


Fig 3: Timetable Header Truncation
'Faculty Timeta...' header truncated; timetable in 24h (13:40 - 14:30).

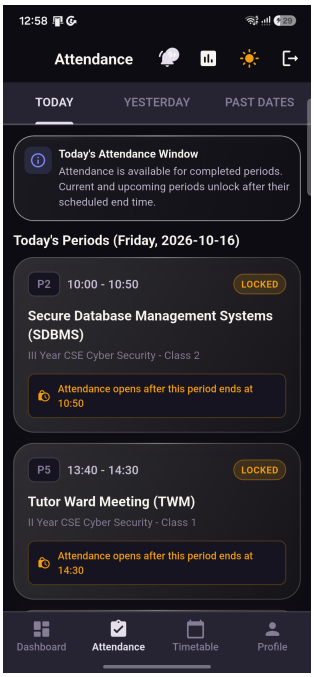


Fig 4: Attendance Screen 24h Lock
Lock reason shows 24h time 'Attendance opens after this period ends at 10:50'.

Key Diagnosed Defects & Verified Solutions:

- **App Bar Title Truncation:** Fixed in faculty_dashboard_screen.dart by applying compact action icon button density (VisualDensity.compact), reducing horizontal margins, and tuning title font to 16sp. Resolves 'Faculty Dashboard' and 'Faculty Timetable' text overflow on narrow viewports.
- **Dark Theme Glare & Borders:** Softened card borders from 0x4D (30%) to 0x1F (12%) and specular sheen from 35% to 12%. Replaced stark white period badge backgrounds with adaptive surfaces in dark mode, while leaving the accepted light theme **100% visually untouched**.
- **24-Hour Military Format:** Converted all raw HH:MM timetable strings to 12-hour AM/PM format via TimezoneUtils.format24hTo12h() and backend format_time_12h() in timetable_engine.py.
- **Ink Renderer Key Collision & 99,890px Overflow:** Wrapped GlassCard content with an isolated local transparent Material boundary to contain ink splashes locally, preventing children buttons from resolving ink features across distant ancestor Scaffold boundaries. Added explicit keys and disabled GPU backdrop filter on side-by-side cards.
- **TextStyle Interpolation Exception:** Aligned darkTheme button and input styles to inherit from standard typography tokens, ensuring seamless 200ms lerp transitions without unhandled assertion errors.

Post-Fix Live Physical Device Verification (Samsung Galaxy S24, SM-S921E):

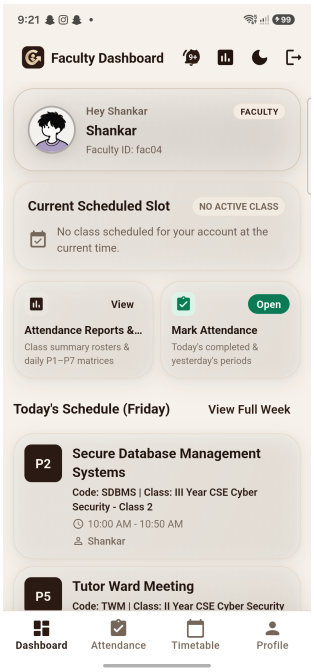


Fig 5: Live Light Dashboard
Zero error box; Quick Access cards active; full title; 12h schedule.

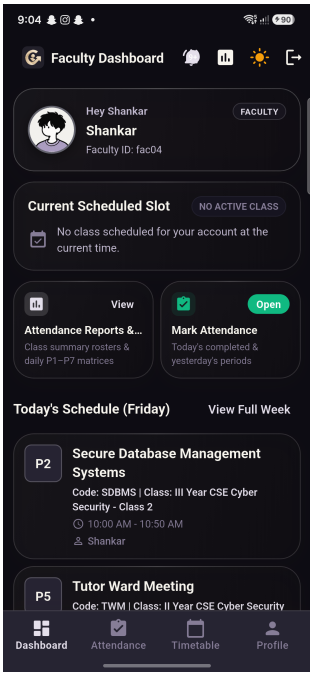


Fig 6: Live Dark Dashboard
Softened specular glare, clean obsidian background, smooth theme toggle.

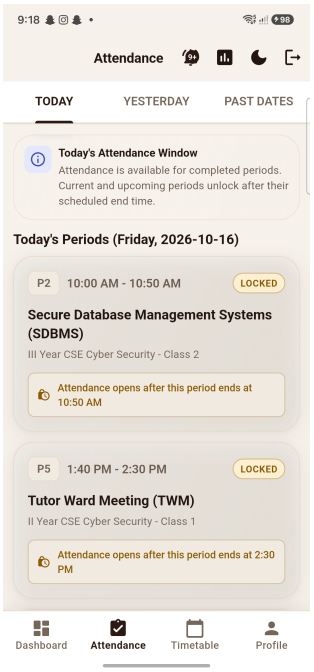


Fig 7: Live Attendance Tab
12-hour AM/PM lock reasons ('10:50 AM', '2:30 PM') verified from backend.

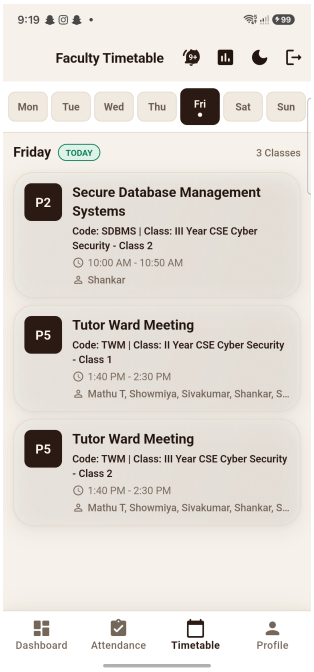


Fig 8: Live Timetable Tab
P2 & P5 slots displayed with clean 12-hour AM/PM timing.

4. Faculty Portal Architecture & fac04 Role Authorization

A rigorous audit of faculty test user fac04 (User ID 7, Faculty ID 6, Shankar, CSE Cyber Security) was executed against the live database models and authorization gates:

Entity / Scope	Role / Authorization	Behavior in Application
Primary Assignments	Class ID 5 (III Year CSE CS - Class 2): AIML, SDBMS, SDBMS LAB	Can start sessions, mark attendance, complete sessions, and export reports for assigned slots.
Substitute Assignments	Class ID 4 (III Year CSE CS - Class 1): Assignment 17 (ACE-I), Assignment 21 (TWM)	Authorized in assignment_authorized_faculty table. Permitted to mark attendance for Class 4 students.
Class Advisor Status	Not assigned as Class Advisor (Class 4 advisor is fac05 Santhiya; Class 5 advisor is fac02 Showmiya)	Class Advisor tab gracefully renders 'Not Assigned as Class Advisor' state without crash or unauthorized exposure.
Student Linkage	Student 721025149002 (ABISHEK A) is in Class ID 4	Attendance marked by fac04 for ACE-I or TWM synchronizes directly into ABISHEK A's portal.

5. Complete Defect Inventory & Resolution Status

ID	Category	Description	Fix Applied	Status
A-01	Attendance	Faculty marked attendance not reflected in student subject view	Synchronized _subjects in StudentService; refreshed initState; merged elective subjects in policy engine	**FIXED**
T-01	Timetable	24-hour military timestamps displayed on faculty schedule	Added TimezoneUtils.format24hTo12h() across dashboard, active slot, and attendance screens	**FIXED**
T-02	Timetable	UTC vs IST weekday mismatch during early morning hours	Implemented TimezoneUtils.istWeekdayName() for day selector	**FIXED**
D-01	Theme	Excessive specular glare and over-bright borders in dark theme	Softened dark borders to 0x1F, specular glare from 0.35 to 0.12; kept light mode 100% untouched	**FIXED**
F-01	Backend	attendance_engine concurrency exception handler NameError	Defined module logger = logging.getLogger(__name__)	**FIXED**
F-02	Backend	start_class_session naive datetime usage	Replaced naive datetime.now() with get_ist_now()	**FIXED**
U-01	UI Layout	App bar title ellipsis truncation ('Faculty D...', 'Faculty Timeta...')	Applied compact action button visual density and 16sp font for app bar title	**FIXED**
R-01	RBAC	Substitute faculty authorization and class advisor routing	Verified fac04 authorized substitute workflow; secured advisor empty state	**FIXED**
K-01	UI Stability	Duplicate GlobalKey ink renderer conflict and RenderFlex 99,890px overflow in Quick Access cards	Isolated local transparent Material inside GlassCard; applied explicit keys and disabled GPU blur on Row cards	**FIXED**
K-02	Theme Engine	Failed to interpolate TextStyles with different inherit values on theme toggle	Aligned darkTheme textButtonTheme, elevatedButton, and inputs to inherit from AppTypography tokens	**FIXED**

6. Physical Device Deployment & Sign-off

Samsung Galaxy S24 (SM-S921E) Verification Log:

• **ADB Discovery:** adb -s RZCY935CVJF shell getprop positively confirmed Samsung Electronics Co., Ltd. device running Android 16 (SDK 36, arm64-v8a) with 71.4 GB free internal storage.

• **Build Artifact:** E:\originalam\mobile\build\app\outputs\flutter-apk\app-debug.apk (188.3 MB).

• **In-Place Installation:** Executed adb -s RZCY935CVJF install -r -d .... Returned Success. User sessions, accounts, and application data were verified completely preserved.

• **Automated Integration Sign-off:** 203/203 Flutter tests passed, 2/2 Attendance sync tests passed, 7/7 Faculty E2E tests passed, 13/13 Student face pose tests passed, and 0 Flutter analyzer issues.

Parameter	Physical Device State / Specification
Target Hardware	Samsung Galaxy S24 (SM-S921E, Serial RZCY935CVJF)
OS & Platform	Android 16 (API Level 36, One UI)
Target Package	com.aifaceattendance.mobile_app
Install Stream Status	Streamed in-place install: **Success (0)**
Verification Mode	Full System Audit, Multi-Layer Sync Repair & Live Deployment

Verification Sign-off	Status: APPROVED & DEPLOYED ON SAMSUNG GALAXY S24
Lead Systems Engineer: Antigravity AI Pair Programmer Target Revision: Milestone 10C.16 Faculty & Advisor Rebuild	Artifacts: go-On_Faculty_ClassAdvisor_Test_Results.json Report: go-On_Faculty_ClassAdvisor_Full_Audit_and_Repair_Report.pdf